

1. (10分) 设 $A = \{\emptyset\}$, $B = P(P(A))$, 判断下列命题是否正确。

(1) $\emptyset \in B, \emptyset \subseteq B$

(2) $\{\emptyset\} \in B, \{\emptyset\} \subseteq B$

(3) $\{\{\emptyset\}\} \in B, \{\{\emptyset\}\} \subseteq B$

(4) $\{\{\emptyset\}, \emptyset\} \subseteq B$

(5) $\{\{\{\emptyset\}\}, \{\emptyset\}\} \subseteq B$

2. (8分) 确定下列集合等式成立的充分必要条件。

(1) $A \cap B = A \cup B$

(2) $(A - C) \cup B = (A \cup B) - C$

(3) $(A - B) \cap (A - C) = \emptyset$

(4) $(A - B) \cap (A - C) = A$

3. (6分) 已知 $A = \{0,1\}$, $B = \{1,2\}$, 求下列集合:

(1) $A \times \{1\} \times B$

(2) $A^2 \times B$

(3) $(B \times A)^2$

4. (12分) 判断下列命题是否正确。

(1) $(A - B) - C = (A - C) - (B - C)$

(2) $\overline{(\bar{A} \cup \bar{B})} \cap \bar{A} = A$

(3) $A - (B \cup C) = (A - B) \cup C$

(4) $(A \cap B) \cup (B - A) \subseteq A$

5. (4分) 设A,B为集合, 证明如果 $(A - B) \cup (B - A) = A \cup B$, 则 $A \cap B = \emptyset$

6. (6分) 设A,B为任意集合, 证明:

(1) $P(A) \cap P(B) = P(A \cap B)$

(2) $P(A) \cup P(B) \subseteq P(A \cup B)$

(3) 针对(2)举一反三, 说明 $P(A) \cup P(B) = P(A \cup B)$ 对某些集合A和B是不成立的。

7. (27分) 设 $\mathbf{R}$ 代表实数集, 针对下列给定函数 f , 判断 $f: \mathbf{R} \rightarrow \mathbf{R}$ 是否为单射或满射。如果不是, 请说明理由。

(1) $f(x) = 2^x$.

(2) $f(x) = \lfloor x \rfloor$.

(3) $f(x) = \begin{cases} (2x-1)/(x-1), & x \neq 1 \\ 0, & x = 1. \end{cases}$

(4) $f(x) = 2^x + x$.

(5) $f(x) = \frac{2x}{x^2+1}$.

(6) $f(x) = x^3 - x^2$.

(7) $f(x) = \begin{cases} x \ln |x|, & x \neq 0 \\ 0, & x = 0. \end{cases}$

(8) $f(x) = \begin{cases} \sqrt{x+1}, & x \in \mathbf{R}^+ \\ 2x, & x \in \mathbf{R} - \mathbf{R}^+. \end{cases}$

(9) $f(x) = \sin x$.

8. (27分) 对于给定的 A, B 和 f , 判断 f 是否为从 A 到 B 的函数 $f: A \rightarrow B$ 。如果是, 说明 f 是否为单射的、满射的、双射的。

(1) $A = \mathbf{Z}, B = \mathbf{N}, f(x) = x^2 + 1$.

(2) $A = \mathbf{N}, B = \mathbf{Q}, f(x) = \frac{1}{x}$.

(3) $A = \mathbf{Z} \times \mathbf{N}, B = \mathbf{Q}, f(\langle x, y \rangle) = \frac{x}{y+1}$.

(4) $A = \{1, 2, 3\}, B = \{p, q, r\}, f = \{\langle 1, q \rangle, \langle 2, q \rangle, \langle 3, q \rangle\}$.

(5) $A = B = \mathbf{N}, f(x) = 2^x$.

(6) $A = B = \mathbf{R} \times \mathbf{R}, f(\langle x, y \rangle) = \langle y+1, x+1 \rangle$.

(7) $A = \mathbf{Z} \times \mathbf{Z}, B = \mathbf{Z}, f(\langle x, y \rangle) = x^2 + 2y^2$.

(8) $A = B = \mathbf{R}, f(x) = \frac{1}{\sqrt{x+1}}$.

(9) $A = \mathbf{N} \times \mathbf{N} \times \mathbf{N}, B = \mathbf{N}, f(\langle x, y, z \rangle) = x + y - z$.